

VLSI Devices

Lecture 22

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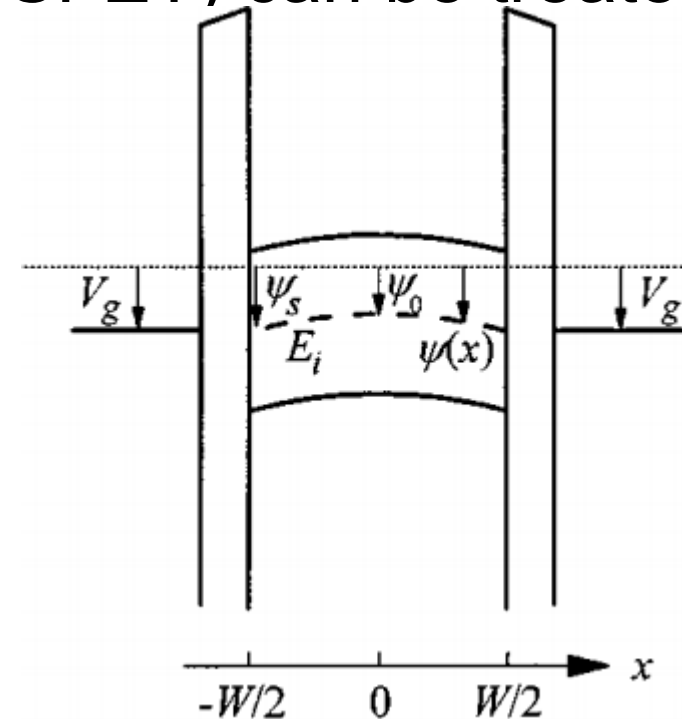
L22

Main difficulty

- Complicated geometry
 - A planar MOSFET (or a double-gate MOSFET) can be treated as a one-dimensional problem.
 - However, multigate MOSFETs?



Cross section of a nanosheet MOSFET channel



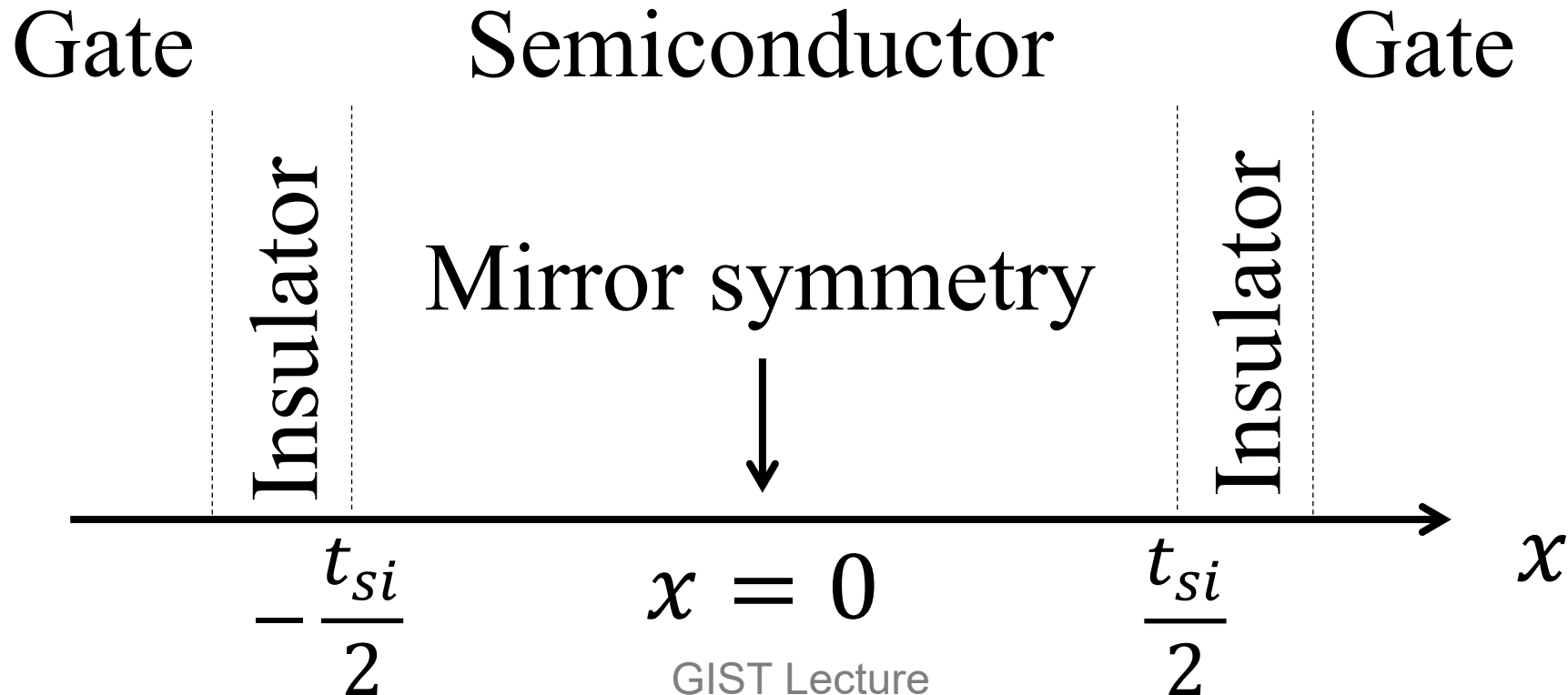
Double-gate MOSFET as a 1D structure (Y. Taur, IEEE EDL, 2000)

Undoped double-gate MOSFET

- Only the mobile charge

$$\frac{d^2\phi}{dx^2} = \frac{q}{\epsilon_{si}} n_i \exp\left(\frac{q(\phi - V)}{k_B T}\right)$$

Taur, Eq. (7.19)



Integrate it.

- We know the trick. A weighting factor of $\frac{d\phi}{dx}$.

$$\frac{1}{2} \frac{d}{dx} \left(\frac{d\phi}{dx} \right)^2 = \frac{q}{\epsilon_{si}} n_i \exp \left(-\frac{qV}{k_B T} \right) \exp \left(\frac{q\phi}{k_B T} \right) \frac{d\phi}{dx}$$

– Integration from 0 to x :

$$\frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 = \frac{q}{\epsilon_{si}} n_i \exp \left(-\frac{qV}{k_B T} \right) \int_{\phi(0)}^{\phi(x)} \exp \left(\frac{q\phi}{k_B T} \right) d\phi$$

– As a result,

$$\frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 = \frac{q}{\epsilon_{si}} n_i \exp \left(-\frac{qV}{k_B T} \right) \frac{k_B T}{q} \left[\exp \left(\frac{q\phi(x)}{k_B T} \right) - \exp \left(\frac{q\phi(0)}{k_B T} \right) \right]$$

Separation of variables

- We assume that $\frac{d\phi}{dx}$ is positive.
 - Then,

$$\frac{d\phi}{\sqrt{\exp\left(\frac{q\phi(x)}{k_B T}\right) - \exp\left(\frac{q\phi(0)}{k_B T}\right)}} = \sqrt{\frac{2k_B T}{\epsilon_{si}} n_i \exp\left(-\frac{qV}{k_B T}\right)} dx$$

- With normalized potential variables, $\tilde{\phi} = \frac{q}{k_B T} (\phi - V)$,

$$\frac{d\tilde{\phi}}{\sqrt{\exp \tilde{\phi}(x) - \exp \tilde{\phi}(0)}} = \sqrt{\frac{2q^2 n_i}{\epsilon_{si} k_B T}} dx$$

Manipulation

- LHS must be integrated.

$$\frac{1}{\sqrt{\exp \tilde{\phi}(0)}} \frac{d\tilde{\phi}}{\sqrt{\exp(\tilde{\phi}(x) - \tilde{\phi}(0)) - 1}}$$

- With a new variable, $u \equiv \tilde{\phi}(x) - \tilde{\phi}(0)$, (u is positive.)

$$\frac{1}{\exp\left(\frac{\tilde{\phi}(0)}{2}\right)} \frac{du}{\sqrt{\exp(u) - 1}}$$

- Then, the main task is

$$\int \frac{du}{\sqrt{\exp(u) - 1}}$$

Introduce $v^2 = \exp(u) - 1$.

- It follows $2v dv = \exp(u) du$.

– The integration becomes

$$\int \frac{du}{\sqrt{\exp(u) - 1}} = \int \frac{\exp(-u) 2v dv}{v} = 2 \int \frac{dv}{v^2 + 1}$$

– It is a known integral.

$$\int \frac{dv}{v^2 + 1} = \arctan v + C$$

– In terms of u ,

$$\int \frac{du}{\sqrt{\exp(u) - 1}} = 2 \arctan \left(\sqrt{\exp(u) - 1} \right) + C$$

Entire LHS and RHS terms

- When u varies from 0 to $\tilde{\phi}(x) - \tilde{\phi}(0)$,

- Then, the entire LHS term becomes

$$\frac{2}{\exp\left(\frac{\tilde{\phi}(0)}{2}\right)} \arctan\left(\sqrt{\exp\left(\tilde{\phi}(x) - \tilde{\phi}(0)\right) - 1}\right)$$

- Of course, the RHS term is

$$x \sqrt{\frac{2q^2 n_i}{\epsilon_{si} k_B T}}$$

Matching them

- By taking tangent,

$$\sqrt{\exp(\tilde{\phi}(x) - \tilde{\phi}(0)) - 1} = \tan \left[x \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T} \exp(\tilde{\phi}(0))} \right]$$

- A simple manipulation yields

$$\exp(\tilde{\phi}(x) - \tilde{\phi}(0)) = \sec^2 \left[x \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T} \exp(\tilde{\phi}(0))} \right]$$

- Now we have

$$\tilde{\phi}(x) = \tilde{\phi}(0) - 2 \log \left[\cos \left(x \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T} \exp(\tilde{\phi}(0))} \right) \right]$$

Introduce β . (It is dimensionless.)

- Following Prof. Taur's book,

$$\frac{2\beta}{t_{si}} = \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T} \exp(\tilde{\phi}(0))}$$

– Then,

$$\exp\left(-\frac{1}{2}\tilde{\phi}(0)\right) = \frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}}$$

– Therefore,

$$\tilde{\phi}(0) = -2 \log \left[\frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}} \right]$$

The solution

- Following Prof. Taur's book,

$$\tilde{\phi}(x) = -2 \log \left[\frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}} \cos \left(\frac{2\beta}{t_{si}} x \right) \right]$$

- Using the electrostatic potential,

$$\phi(x) = V - \frac{2k_B T}{q} \log \left[\frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}} \cos \left(\frac{2\beta}{t_{si}} x \right) \right] \quad \text{Taur, Eq. (7.20)}$$

- At the interface, $x = \frac{t_{si}}{2}$, the surface electric field becomes

$$\phi \left(\frac{t_{si}}{2} \right) = V - \frac{2k_B T}{q} \log \left[\frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}} \cos(\beta) \right]$$

Surface electric field

- Calculate $\frac{d\phi}{dx}$.

$$\frac{d\phi}{dx} = \frac{2k_B T}{q} \frac{2\beta}{t_{si}} \tan\left(\frac{2\beta}{t_{si}} x\right)$$

- At the interface, $x = \frac{t_{si}}{2}$, the surface electric field becomes

$$\frac{2k_B T}{q} \frac{2\beta}{t_{si}} \tan(\beta)$$

- Therefore, the potential at the gate contact is given as

$$V_{gs} - \Phi_{mi}$$

Taur, Eq. (7.22)

$$= V - \frac{2k_B T}{q} \log \left[\frac{t_{si}}{2\beta} \sqrt{\frac{q^2 n_i}{2\epsilon_{si} k_B T}} \cos(\beta) \right] + \frac{2k_B T}{q} \frac{\epsilon_{si}}{\epsilon_{ox}} \frac{2t_{ox}}{t_{si}} \beta \tan(\beta)$$

Homework#21

- Due: 08:00 on June 1
- Submit a PDF report through the GIST LMS system.
 - Draw the electrostatic potential of Taur, Eq. (7.20).
 - Silicon thickness is 5 nm.
 - You may assume $V = 0$, 300 K, $t_{ox} = 0.7$ nm, and $\Phi_{mi} = -0.51$ V.
 - By varying β , calculate the surface electric field and the electrostatic potential at the gate contact.

Thank you!